

1. what is Primary and Secondary Data collection?

Primary and secondary data collection are fundamental concepts in both general intelligence and the specific fields of **Cyber Intelligence** and **Counter Intelligence**.

The distinction lies in the origin and purpose of the data.¹

Primary Data Collection

Primary data is **first-hand, original information** collected directly by the intelligence team or researcher specifically for the current intelligence requirement or investigation.² It's new data gathered to answer a specific question.³

Key Characteristics:

- **Originality:** The data is newly generated.
- **Purpose-Driven:** Collected to address a specific **Key Intelligence Question (KIQ)** or objective.⁴
- **Time & Cost:** Often more **time-consuming** and **expensive** to collect.⁵
- **Relevance:** Provides the highest level of relevance and control over data quality.⁶

Examples in Cyber and Counter Intelligence:

- **Interviews:** Debriefings of personnel, interviews with subjects or witnesses of a cyber-attack, or direct contact with Subject Matter Experts (SMEs).⁷
- **Active Technical Collection:**
 - Running a **penetration test** (pen-test) against an organization's network to generate specific vulnerability data.⁸
 - Deploying a **honeypot** or **honeynet** to directly lure attackers and observe their **Tactics, Techniques, and Procedures (TTPs)** in real-time.
 - Conducting **live malware analysis** in a sandbox environment to gather Indicators of Compromise (IoCs) for a specific, current threat.
- **Internal Telemetry:** Logs, network traffic monitoring, and Endpoint Detection and Response (EDR) data collected directly from an organization's own systems.⁹



Secondary Data Collection

Secondary data is **second-hand, existing information** that has already been collected, processed, and published (or held) by someone else for a different purpose.¹⁰ This data is leveraged by the intelligence team to provide context, background, and a broader view.

Key Characteristics:

- **Existing:** The data already exists in a collected and often analyzed form.¹¹
- **General Purpose:** Was originally gathered for another reason; its utility is repurposed.¹²
- **Time & Cost:** Generally **faster** and **less expensive** to acquire.¹³
- **Scope:** Provides a **broad overview** of the threat landscape or adversary.¹⁴

Examples in Cyber and Counter Intelligence:

- **Open Source Intelligence (OSINT):**
 - **Public Reports:** Threat intelligence feeds, vendor reports, academic papers, and government publications (e.g., CISA alerts).¹⁵
 - **Public Records:** WHOIS records for domain registration, public court filings, company financial statements.¹⁶
 - **Media:** News articles, social media posts, and forum discussions (including those on the deep/dark web) that have been previously posted.¹⁷
- **Historical Data:** Past incident reports, logs from previous security events, and historical **Indicators of Compromise (IoCs)** databases.
- **Commercial Feeds:** Purchased, pre-aggregated data from commercial threat intelligence providers.¹⁸



Combined Use in Intelligence

In practice, effective Cyber and Counter Intelligence relies on an **integration of both primary and secondary data** .

- The intelligence cycle usually **starts with secondary research** to understand the general landscape and identify potential threats.¹⁹

- **Primary research** is then used to **validate signals** found in secondary data, **fill in gaps** where existing information is insufficient, or conduct a **deep dive** on a specific, high-priority threat identified through monitoring.²⁰ For example, a secondary source might identify a new threat actor (what they are doing), and primary collection (e.g., a honeypot) might be used to understand *how* they execute their attack.
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Types of Intelligence Sources

Source are osint, humint, sigint

Cyber Intelligence Types

Strategic Cyber Intelligence –

- Strategic threat intelligence is high-level intelligence about the global threat landscape and an organization's place within it.
- Strategic threat intelligence gives decision-makers outside of IT, such as CEOs and other executives, an understanding of the cyberthreats their organizations face.
- Strategic threat intelligence usually focuses on issues such as geopolitical nation-state or geopolitical alliance situations, cyberthreat trends in a particular industry, or how or why certain of the organization's strategic assets may be targeted.
- Stakeholders use strategic threat intelligence to align broader organizational risk management strategies and investments with the cyberthreat landscape.

Operational Cyber Intelligence –

Operational threat intelligence helps organizations anticipate and prevent future attacks.

- It is sometimes called 'technical threat intelligence' because it details the TTPs and behaviors of known threat actors—for example, the attack vectors they use, the vulnerabilities they exploit and the assets they target.
- CISOs, CIOs and other information security decision-makers use operational threat intelligence to identify threat actors who are likely to attack their organizations

and respond with security controls and other actions aimed specifically at thwarting their attacks.

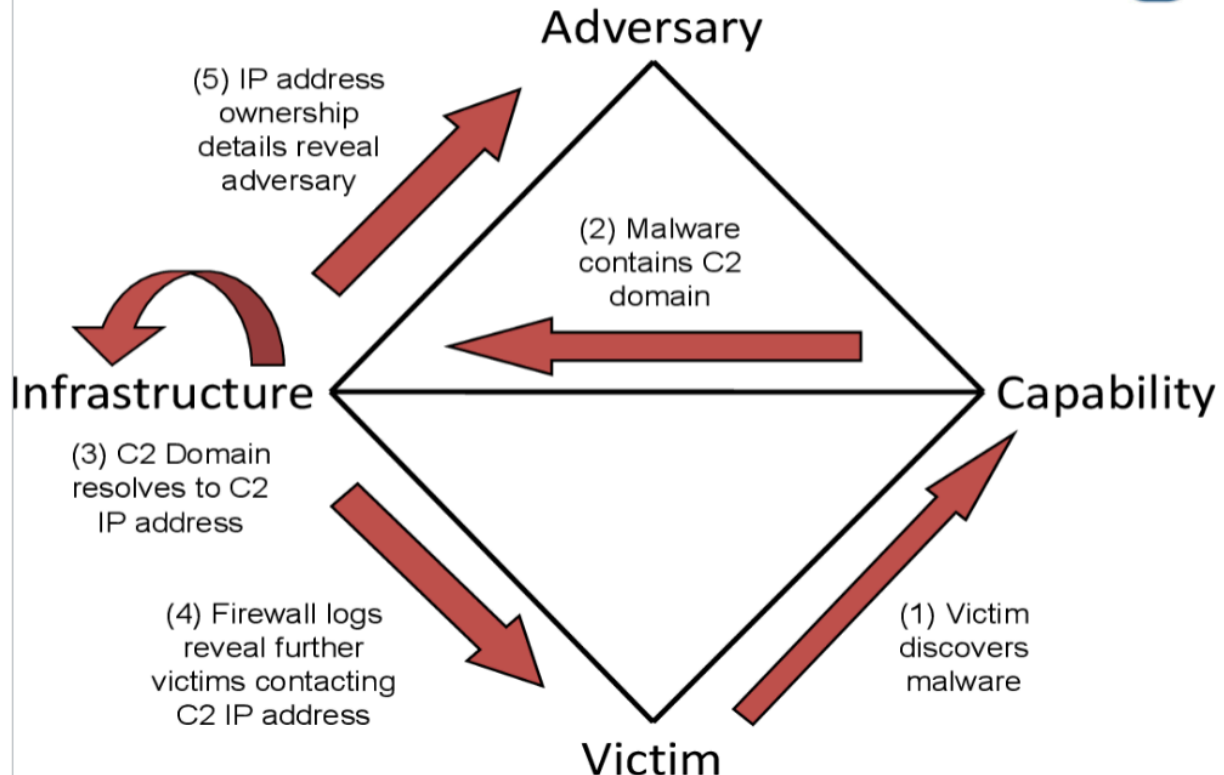
Technical Cyber Intelligence –

- Technical indicators related to an actor's tools, malware, infrastructures, that are used to conduct their activities.
- Focuses on IOC's and IOA's and malware analysis.
- Provides specific hashes and details pertaining to a specific attack.
- Informs defenders , threat detection and response managers, incident responders, and technical teams.
- Feeds & Reports include malicious hashes, registry mods, specific malware identifiers, file names/hashes.

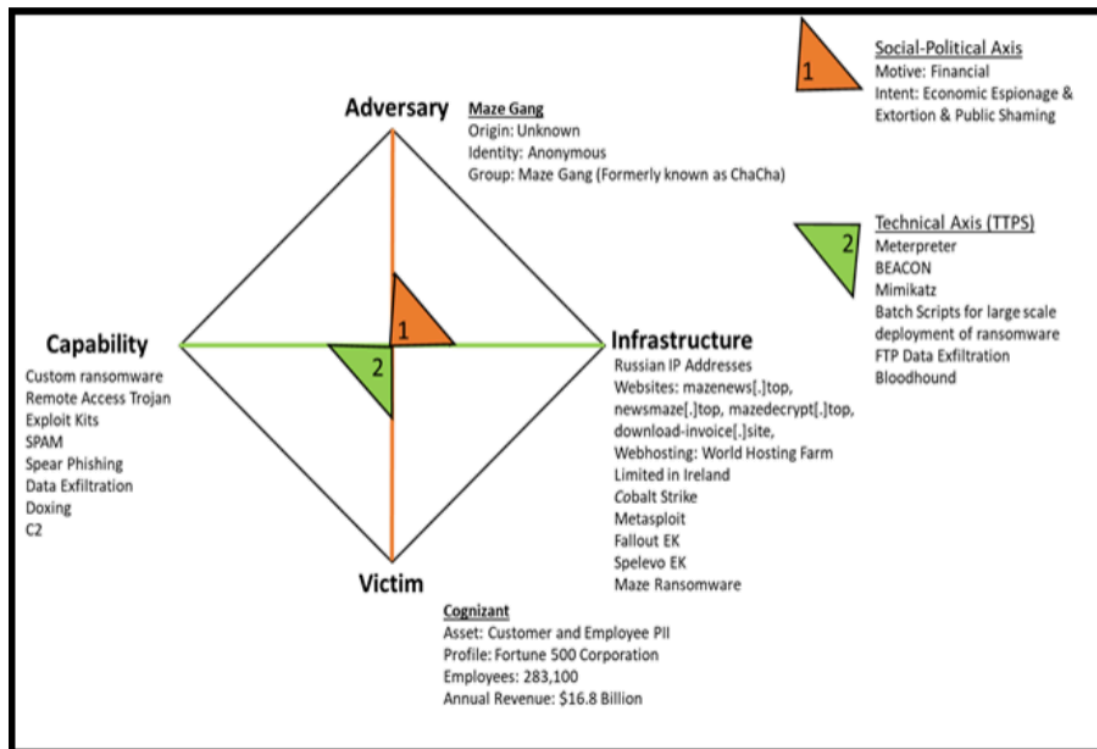
Tactical Intelligence –

- Tactical threat intelligence is used by the security operations center (SOC) to detect and respond to cyberattacks in progress.
- It typically focuses on common IoC's—for example, IP addresses associated with command and control servers, file hashes related to known malware and ransomware attacks or email subject lines associated with phishing attacks.
- In addition to helping incident response teams filter out false positives and intercept genuine attacks, tactical threat intelligence is also used by threat-hunting teams to track down advanced persistent threats (APTs) and other active but hidden attackers.

The Diamond Model



The Diamond Model



How Does the Diamond Model Work?

There are four main components of the Diamond Model of Intrusion:

- Adversary: The attacker or group responsible for a cyber incident.
- Infrastructure: The technical resources or assets the adversary uses during the attack (e.g., servers, domains, and IP addresses).
- Capability: A method, tool, or technique the adversary uses during the attack (e.g., malware or exploits).
- Victim: The individual or organization the adversary targets during the attack.

There are also various relationships between these components, including:

- Adversary-victim: The interaction between the attacker and target. This relationship concerns questions such as why the attacker selected this target and the attacker's motivations and objectives.
- Adversary infrastructure: The attacker uses various technical resources and assets. This relationship concerns how the attacker establishes and maintains its cyber operations.
- Victim-infrastructure: The target's connection to the attacker's technical resources. This relationship concerns the attacker's use of various channels, methods, and vectors against the target.
- Victim-capability: The target's connection to the attacker's tools and techniques. This relationship concerns specific tactics and attack signatures used against the target.

How Does the Diamond Model Work?

What Are the Benefits of Using the Diamond Model?

The Diamond Model of Intrusion Analysis offers advantages such as:

- Holistic understanding: The Diamond Model examines the technical aspects of a cyberattack and the human and organizational aspects (in the form of the adversary and victim).
- Structured analysis: The Diamond Model provides a clear, organized way for cybersecurity experts to structure and process data relating to cyber threats and attacks, making it easier to collaborate and share information.
- Incident response and threat intelligence: The Diamond Model offers benefits both for threat intelligence (before an attack) and incident response (after an attack), helping analysts collect and analyze valuable data.

Technical Intelligence

•This includes all the technical aspects of cyber threats, such as malware, exploits, hacking tools, attacker infrastructure (domains and IP addresses), and TTPs. You will commonly collect this from external CTI feeds (commercial and open source) and your own internal log data.

Internal vs External Collection Sources

1. Internal Sources – Network element logs and records of past incident responses. Intrusion analysis data and malware analysis data, domain information, TLS/SSL certificates.
2. External Sources – External malware analysis and online sandbox tools; technical blogs and magazines, the dark web, open sources, and counterintelligence data, to collect information about adversaries' signatures.
3. Collected data is placed into lists of Indicators of Compromise (IOC's) that include domain information, IP addresses, SSL/TLS certificates information, file hashes, network scanning information, vulnerability assessment information, social media news (in raw format), email addresses, email senders, email links, and attachments.

Threat hunting

Threat hunting is a proactive approach to cybersecurity in which security teams discover threats before they attack the systems.

With threat hunting, organizations can find even more sophisticated threats that can go undetected by existing security measures, such as fileless malware.

It begins with malicious activity triggers and then proceeds with analysis and threat resolution phases.

Security teams use a variety of tools and technologies to automate threat hunting.

- Sqrrl has developed a Threat Hunting Loop consisting of four stages that define an effective hunting approach.
- The goal of a hunt team should be to get through the loop as quickly and effectively as possible.
- The more efficiently you can iterate, the more you can automate new processes and move on to finding new threats.

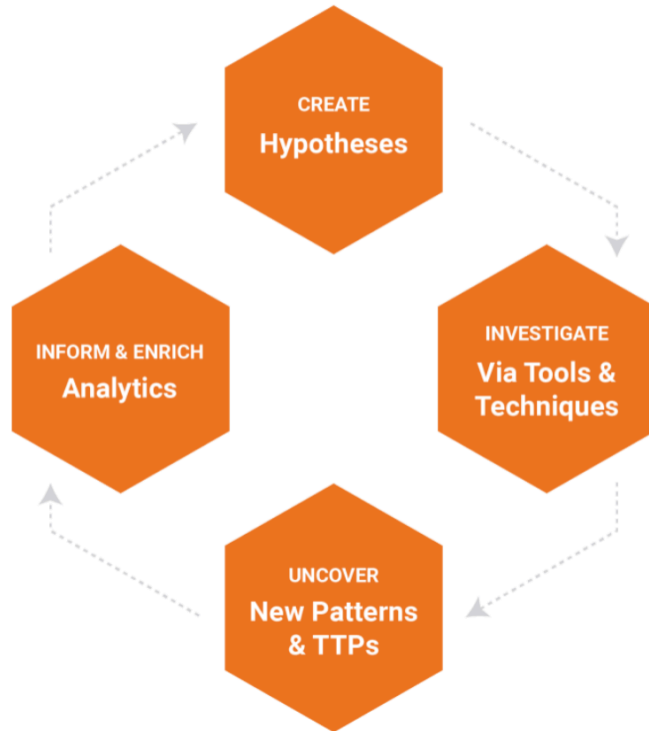
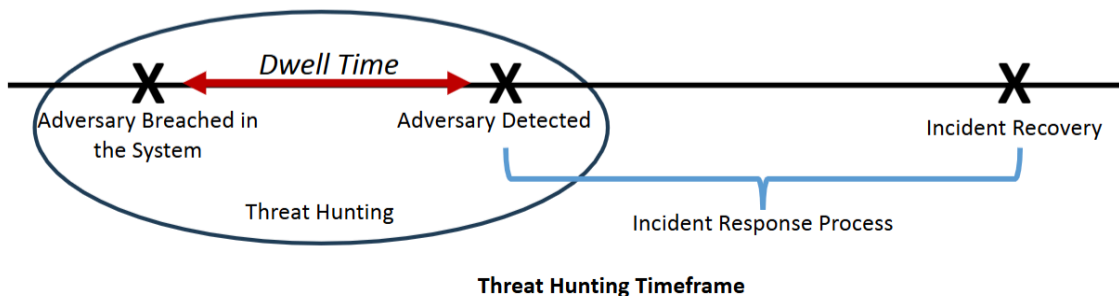


FIGURE 2. THE HUNTING LOOP



To start threat hunting, use tools such as:

1. Managed detection and response (MDR)
2. SIEM
3. Security analytics
4. Endpoint detection and response (EDR)

Domains and IPs

Domains and Internet Protocol (IP) addresses are the second most common intelligence collection source.

intelligence requirement

Any subject, general or specific, upon which there is a need for the collection of information, or the production of intelligence.

- This stakeholder could be a member of the security operations team who is triaging an incident or an executive who needs intelligence to make an informed **strategic decision** about future security investments.
- Intelligence requirements should be simple, focused, and **support a single decision** that a **key stakeholder** needs to make.

Prioritize Resource Allocation and Efforts

- Establishing clear intelligence requirements will help you allocate resources effectively. For instance, if you need to collect data about domains to fulfill an intelligence requirement, you can focus your collection on domain-related data sources (e.g., network flow logs, WAF logs, external data sources like Censys, etc.).
- This enables you to save time and money by investing in data sources relevant to intelligence needs rather than blindly collecting data.
- Intelligence requirements also highlight your organization's current security capabilities to management.
- Executives can see what the organization prioritizes to defend itself from cyber threats, what technologies are being used, and the value these technologies deliver.
- An intelligence requirement that is not being addressed or a new one that is created might need a new tool, technology, or data source.
- Management can judge based on existing intelligence requirements if they want to make this investment.

- **Priority Intelligence Requirements (PIR)**: These are intelligence requirements that are **mission critical**.
- They are **key to your organization's security success and will change based on the need to address immediate threats or situations**.
- For example, your team may receive a **PIR** from the incident response team regarding additional IOCs to search for based on the latest version of Qakbot malware.
- **PIRs should be bound to specific events or times, focusing only on critical needs.**

Best data collection procedure

Ensure data is reliable and accurate

Define clear objectives

Regularly review & update collection procedure

Intelligence requirement prioritize and risks?

Singular atomic Decision centric timeline

Risk prioritization factors?

* Likelihood of attack , severity of attack , cost of remediation , ease of remediation, regularity compliance

Prioritize threats

Relevance magnitude evolution context